

Spectral Theorem for Compact Self-adjoint operators

→ Let A be a compact, self-adjoint op. on an ∞ -dim Hilbert Space H . Then H has a complete orthonormal system $\{v_1, v_2, \dots\}$ consisting of eigenvectors of A . For every $\psi \in H$

$$(A\psi) = \sum_{n=1}^{\infty} \lambda_n \langle \psi | v_n \rangle v_n$$

where λ_n is the eigenvalue corresponding to v_n .

Hilbert-Schmidt Theorem:

→ For A as above, $\exists$ a system of eigen-vectors (u_n) corresponding to eigenvalues (λ_n) s.t. ALL $\psi \in H$ has a unique rep. of the form:

$$\psi = \sum_{n=1}^{\infty} \alpha_n u_n + \eta \quad \left| \quad \begin{array}{l} \alpha_n \in \mathbb{C} \\ \eta \in N(A) \end{array} \right.$$

Null Space $\rightarrow$

→ Translating this to "physics" we are basically allowed to have the wave function, an element of the Hilbert Space $L^2(\mathbb{R})$, to be represented in a multitude of ways. This could be in position space

$$\langle x | \psi \rangle = \psi(x)$$

momentum space

$$\langle p | \psi \rangle = \psi(p)$$

"spin" states, etc...

Dirac Notation: (Energy eigenstates)

$$|n\rangle = |\psi_n\rangle = |E_n\rangle = |(n + \frac{1}{2})\hbar\omega\rangle$$

$$|0\rangle = |\psi_0\rangle = |E_0\rangle = |\frac{1}{2}\hbar\omega\rangle$$

$$\psi_n(x) = \langle x | n \rangle$$

$\Rightarrow$ Will often see $|n\rangle \doteq \psi_n(x)$

(i) Orthogonality:

$$\langle \psi_n | \psi_m \rangle = \delta_{nm} = \int_{-\infty}^{\infty} dx \psi_n^*(x) \psi_m(x)$$

$$\langle n, m \rangle = \delta_{nm}$$

(ii) Completeness relation:

$$\langle \psi_n, \psi_n \rangle = \int_{-\infty}^{\infty} dx \psi_n^*(x) \psi(x)$$

$$\text{but } \psi_n(x) = \langle x | n \rangle$$

$$\psi_n^*(x) = \langle n | x \rangle = (\langle x | n \rangle)^*$$

$$\begin{aligned} \therefore &= \int_{-\infty}^{\infty} dx \langle n | x \rangle \langle x | n \rangle = \langle n | \left(\int_{-\infty}^{\infty} dx |x\rangle \langle x| \right) |n\rangle \\ &= \langle n | n \rangle = 1 \end{aligned}$$

Thus, we conclude

$$I = \int_{-\infty}^{\infty} dx |x\rangle \langle x|$$

* Completeness of expansions in $|n\rangle$ means

$$|\psi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$$

$$\langle m | \psi \rangle = \sum_{n=0}^{\infty} c_n \langle m, n \rangle$$

$$\langle m | \psi \rangle = c_m$$

$$\Rightarrow |\psi\rangle = \sum_{n=0}^{\infty} \langle n | \psi \rangle |n\rangle = \sum_{n=0}^{\infty} (|n\rangle \langle n|) |\psi\rangle$$

$$I |\psi\rangle = \left(\sum_{n=0}^{\infty} |n\rangle \langle n| \right) |\psi\rangle$$

• This is just the "physics" way of writing the Hilbert-Schmidt Theorem.

$$\hat{I} = \sum_{n=0}^{\infty} |n\rangle \langle n|$$

Wave f^u :

$$\langle \psi | \psi \rangle = \sum_{n=0}^{\infty} c_n \langle \psi | n \rangle$$

$$= \int_{-\infty}^{\infty} dx \psi^*(x) \psi(x) = \sum_{n=0}^{\infty} c_n \int_{-\infty}^{\infty} \langle \psi | x \rangle \langle x | n \rangle dx$$

$$= \int_{-\infty}^{\infty} dx \psi^*(x) \sum_{n=0}^{\infty} c_n \langle x | n \rangle$$

$$\psi(x) = \sum_{n=0}^{\infty} c_n \langle x | n \rangle \rightarrow \phi_n(x)$$

$$\begin{aligned} c_n &= \int_{-\infty}^{\infty} dx \langle n | x \rangle \langle x | \psi \rangle \\ &= \int_{-\infty}^{\infty} dx \phi_n^*(x) \psi(x) \end{aligned}$$

$$1 = \langle \psi | \psi \rangle = \sum_n \langle \psi | n \rangle \langle n | \psi \rangle = \sum_{n=0}^{\infty} |c_n|^2$$

* The probability that the state $|\psi\rangle$ is measured to be in $|n\rangle$ (i.e. that it has energy E_n)

$$P_{E_n} = |\langle n | \psi \rangle|^2 = |c_n|^2$$